Exam: Business Intelligence; EDV No.:282183; Type of exam: PK
Examiner: Prof. Dr.-Ing. Dr.-Ing. habil. Javier Villalba-Diez

Last name, first name: ,

Student ID number:

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Date: 20250709; Location: Bildungscampus; Working time: 120 min.; Allowed aids: non-programmable calculator. Total score: 100 points. FURTHER EXPLANATIONS [1]. IMPORTANT: YOU MUST SHOW ALL YOUR CALCULATIONS!

1. **Ex (20 Points):**
   (a) Consider the undirected KPI dependency graph $G = (0-1,\ 0-2,\ 1-2,\ 2-3,\ 3-4,\ 3-5,\ 4-5,\ 5-6,\ 6-u,\ 0-v)$ with $u = (a+1)$ and $v = (a+3)$. Draw G, list the degree of each node, and give the degree distribution. (4 Points)
   (b) Compute the shortest path distances between all unordered node pairs and the average path length (APL) of G. (6 Points)
   (c) Compute the local clustering coefficient C_i for every node and the average clustering coefficient $\bar{C}$. (6 Points)
   (d) Write the adjacency matrix A and the Laplacian matrix $L = D - A$ (node order $0, 1, 2, 3, 4, 5, 6$). (4 Points)

2. **Ex (20 Points): Decision Tree (CART) — first split, step by step**
   (a) **Root node (all observations).** Using the full dataset (all 6 customers), determine the class counts $n_1 = \#\{Y = 1\}$ and $n_0 = \#\{Y = 0\}$. Then compute the Gini impurity of the root node. (4 Points)

ID	X_1 (usage h/week)	X_2 (tickets)	Y (churn)
1	$2 + 0.1a$	1	1
2	$3 + 0.1b$	0	1
3	$4 + 0.1c$	0	0
4	$5 + 0.1d$	1	0
5	$7 + 0.1e$	2	0
6	$8 + 0.1f$	2	0

   (b) **Candidate split search on X_1.** Consider the candidate thresholds

$$t \in \{2.5,\ 3.5,\ 4.5,\ 6.0,\ 7.5\}.$$

   For each threshold t, do the following:
      i. Form the two groups $L(t) = \{X_1 \leq t\}$ and $R(t) = \{X_1 > t\}$, and write down their sizes $|L(t)|$ and $|R(t)|$. (2 Points)
      ii. For each group, compute the class counts (n_1, n_0) and the corresponding Gini impurity. (4 Points)
      iii. Compute the *weighted* Gini impurity of the split:

$$G_{\text{split}}(t) = \frac{|L(t)|}{6}\, G(L(t)) + \frac{|R(t)|}{6}\, G(R(t)).$$

   (2 Points)
   (c) **Candidate split on X_2.** Split the data into $L = \{X_2 \leq 0.5\}$ and $R = \{X_2 > 0.5\}$. Compute the Gini impurity of each side and then the weighted Gini impurity of this split. (4 Points)
   (d) **Choose the CART root split.** Compare all candidate splits from items 2 and 3 and select the split with the smallest weighted Gini impurity. Report:
      i. the selected feature and threshold,
      ii. the predicted class on the left leaf and on the right leaf (majority class on each side; if tied, state a tie-break rule).

   (4 Points)

3. **Ex (20 Points):** A product team models revenue y (in k€) as a linear function of daily active users x (in k users):

$$\hat{y} = \beta_0 + \beta_1 x.$$

   Historical observations:

Day	1	2	3	4	5	6
x	1	2	3	4	5	6
y	$4 + a$	$7 + b$	$10 + c$	$13 + d$	$16 + e$	$19 + f$

[1] Please mark every sheet with your name, student ID number, and degree program. Leave a 3 cm margin on each page. Use only permanent (document-proof) pens (no pencils). For module examinations with multiple examiners: please use separate exam sheets for each examiner. Use only the provided exam paper. Please write only on one side of each page. No questions are permitted during the exam. Only legible exams will be graded!

(a) Fit the OLS regression line by computing β_1 and β_0 using

$$\beta_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}, \qquad \beta_0 = \bar{y} - \beta_1 \bar{x}.$$

(10 Points)

(b) Interpret β_1 in business terms (units and meaning). (5 Points)

(c) Forecast $\hat{y}$ for $x^* = 7 + 0.1a$ and show the computation. (5 Points)

4. **Ex (20 Points):** A BI team models churn probability $p(x)$ as a function of weekly usage x (hours/week) using logistic regression:

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}.$$

The model is assumed to match these two points exactly:

$$p(2 + 0.1c) = 0.75 + 0.01a, \qquad p(8 + 0.1d) = 0.25 + 0.01b.$$

(a) Write the two equations in β_0, β_1 using the logit link

$$\operatorname{logit}(p) = \ln\left(\frac{p}{1-p}\right).$$

(6 Points)

(b) Solve for β_0 and β_1 (numerical approximation is sufficient). (8 Points)

(c) Compute $p(x^*)$ for $x^* = 5 + 0.1e$ and interpret the result (risk statement). (6 Points)

5. **Ex (20 Points):** Given are six KPIs $K_1, \ldots, K_6$ with ten observed values each (e.g., ten weeks). The organization has four levels: Level 1 (highest) to Level 4 (lowest).

	1	2	3	4	5	6	7	8	9	10
K_1	$18 + a$	$17 + b$	$19 + c$	$16 + d$	$15 + e$	$14 + f$	$13 + a$	$12 + b$	$11 + c$	$10 + d$
K_2	$60 + a$	$62 + b$	$61 + c$	$63 + d$	$64 + e$	$65 + f$	$66 + a$	$64 + b$	$63 + c$	$62 + d$
K_3	$4 + 0.1a$	$3 + 0.1b$	$5 + 0.1c$	$4 + 0.1d$	$3 + 0.1e$	$2 + 0.1f$	$3 + 0.1a$	$4 + 0.1b$	$3 + 0.1c$	$2 + 0.1d$
K_4	$88 + a$	$89 + b$	$90 + c$	$91 + d$	$92 + e$	$93 + f$	$92 + a$	$91 + b$	$90 + c$	$89 + d$
K_5	$6 + 0.1a$	$6 + 0.1b$	$7 + 0.1c$	$7 + 0.1d$	$8 + 0.1e$	$8 + 0.1f$	$7 + 0.1a$	$7 + 0.1b$	$6 + 0.1c$	$6 + 0.1d$
K_6	$20 + a$	$18 + b$	$19 + c$	$17 + d$	$16 + e$	$15 + f$	$14 + a$	$13 + b$	$12 + c$	$11 + d$

K_1 = Lead Time (days), K_2 = OEE (%), K_3 = Scrap Rate (%), K_4 = On-Time Delivery (%), K_5 = Inventory Turns (1/year), K_6 = Rework Hours (h/week).

(a) Explain the CPDnA cycle of Lead Time (K_1) being reported from Level 2 to Level 3. Your explanation must include: (i) which information is checked and by whom, (ii) how planning is performed, (iii) how execution is ensured, and (iv) how actions/standardization and escalation are handled. (10 Points)

(b) Describe the Hoshin Kanri tree for the organization (Level 1 to Level 4) and its management process. Include at least one objective per level and explain how KPIs $K_1, \ldots, K_6$ are used in catchball and review routines. (10 Points)

Answer Sheet (to be scanned): Business Intelligence; EDV No.:282183

Last name, first name: ____

Student ID number:

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Write your **final answers** in the boxes below (use additional paper for scratch work if needed).

Ex 1(a)	
Ex 1(b)	
Ex 1(c)	
Ex 1(d)	
Ex 2(a)	
Ex 2(b)	
Ex 2(c)	
Ex 2(d)	
Ex 3(a)	
Ex 3(b)	
Ex 3(c)	
Ex 4(a)	
Ex 4(b)	
Ex 4(c)	
Ex 5(a)	
Ex 5(b)	